

Unique Elements Function

Write a function `unique_elements(lst)` that returns a list containing only the unique elements from the given list (order should be preserved)

```
def unique_elements(lst):
    element = set()
    result = []
    for item in lst:
        if item not in element:
            result.append(item)
            element.add(item)
    return result
```

```
data = [1, 2, 2, 3, 4, 1, 5]
print(unique_elements(data))
[1, 2, 3, 4, 5]
List Rotation
```

Write a function `rotate_list(lst, k)` that rotates the elements of a list to the right by `k` positions.

```
def rotate_list(lst, k):
    if not lst:
        return []
    k = k % len(lst)
    return lst[-k:] + lst[:-k]
```

```
print(rotate_list([1, 2, 3, 4, 5], 2))
[4, 5, 1, 2, 3]
```

Find Longest Word

Write a function `longest_word(sentence)` that returns the longest word from a given sentence.

```
def longest_word(sentence):
    words = sentence.split()
    return max(words, key=len)
```

```
print(longest_word("Python is an amazing programming language"))
programming
```

Sum of Digits Function

Write a function `sum_of_digits(num)` that calculates and returns the sum of digits of a number.

```
def sum_of_digits(num):
    num = abs(num)
    return sum(int(digit) for digit in str(num))
```

```
print(sum_of_digits(12345))
15
```

Character Frequency Counter

Write a function `char_frequency(s)` that returns a dictionary with characters as keys and their frequency as values.

```
def char_frequency(s):
    freq = {}
    for char in s:
        freq[char] = freq.get(char, 0) + 1
    return freq
print(char_frequency("hello"))
{'h': 1, 'e': 1, 'l': 2, 'o': 1}
```

Loops

Numbers Divisible by 3 or 5

Print all numbers from 1 to 100 that are divisible by 3 or 5 but not both.

```
for i in range(1, 101):
    if i % 3 == 0 or i % 5 == 0:
        print(i, end=" ")
3 5 6 9 10 12 15 18 20 21 24 25 27 30 33 35 36 39 40 42 45 48 50 51 54 55 57 60 63 65 66 69 70 72 75 78 80 81 84 85 87 90 93 95 96 99 100
```

Reverse Words in a Sentence

Without using Python's built-in `split()` and `reverse()` in one line, reverse the order of words in a sentence using loops

```
sentence = "Python is fun"
word, words = "", []
```

```

for ch in sentence + " ":
    if ch != " ":
        word += ch
    else:
        words.append(word)
        word = ""
print(" ".join(words[:-1]))
fun is Python
Star Diamond Pattern
For n = 5, print a diamond pattern:

```

```

*
***

n=5
for i in range(1, n + 1, 2):
    spaces = (n - i) // 2
    stars = i
    print(" " * spaces + "*" * stars)
*
***
*****

```

Count Consonants in a String

Write a program to count how many consonants are in a given string

```

text = "hello world"
vowels = "aeiouAEIOU"
count = 0

for ch in text:
    if ch.isalpha() and ch not in vowels:
        count += 1

```

```

print(count)
7
Number Guessing Game

```

Create a loop that keeps asking the user to guess a secret number until they get it right. Guess the number: 5
 Wrong, try again.
 Guess the number: 8
 Correct! You guessed it.

```

secret_number = 8
guess = None

while guess != secret_number:
    guess = int(input("Guess the number: "))
    if guess != secret_number:
        print("Wrong, try again.")
    else:
        print("Correct! You guessed it.")
        Correct! You guessed it.

```

